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DYNAMIC INTERACTIONS BETWEEN ACCOMMODATION AND CONVERGENCE ARE VELOCITY SENSITIVE

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Abstract—Aftereffects of accommodation and vergence occur following approximately 1 min of adaptation to lenses and prisms respectively. This observation can be interpreted to mean that accommodation and vergence responses have phasic and tonic components. We have examined the role that these proposed subcomponents play in mutual interactions between accommodation and vergence. Both accommodative vergence (AV) and vergence accommodation (VA) were unresponsive to low temporal frequency variations (<0.1 Hz) in defocus and disparity respectively. However, both AV and VA were responsive to higher temporal frequency stimuli (up to 0.5 Hz). When negative feedback to the stimulated system was cancelled electronically, both AV and VA become responsive to low temporal frequency stimuli. The ratio or gain of accommodative vergence/accommodation (AC/A) and vergence accommodation/vergence (CA/C) increased nonlinearly with stimulus amplitude. Vergence aftereffects resulted from stimulation of AV and accommodative aftereffects resulted from stimulation of VA. These results are interpreted to mean there could be a complementary relationship between the amplitude of the AC/A ratio and proposed tonic adaptation of accommodation, and between the amplitude of the CA/C ratio and proposed tonic adaptation of vergence. A low saturation limit or stimulus window for tonic adaptation may account for the amplitude dependent nonlinearities of the AC/A and CA/C ratios.

Accommodative vergence Vergence accommodation Tonic vergence Tonic accommodation Adaptation Dynamic interaction Velocity sensitive

INTRODUCTION

Accommodation and vergence are two visual-motor systems that adjust the optics and binocular alignment of the eyes in response to changing viewing distance. The dynamic responses of both systems have been characterized as rapid phasic and slow tonic phases (Schor, 1979; Schor *et al.*, 1984). The tonic phase is indicated by aftereffects of disparity vergence responses on the phoria (open-loop vergence) (Schor, 1979), and aftereffects of accommodative responses upon the resting focus of accommodation (Baker *et al.*, 1983; Ebenholtz, 1983, Schor *et al.*, 1984; Schor *et al.*, 1986). The interaction between proposed phasic and tonic responses has been described in terms of a feedback control system (Schor, 1979) because the vergence and accommodative responses continue until their respective errors (disparity and defocus) are reduced to unnoticeable amplitudes. Feedback could allow some proportion of the phasic response of disparity vergence or optical reflex accommodation to be replaced gradually by postulated adaptable tonic vergence and tonic accommodation respectively.

Another potential source of innervation for vergence comes from accommodation (accommodative vergence) (Mueller, 1826). Similarly, accommodation receives innervation resulting from vergence stimulation (vergence accommodation) (Fincham and Walton, 1957). While it is clear that accommodative vergence is driven by the innervation to accommodation (Fry, 1937; Morgan and Peters, 1931; Alpern *et al.*, 1961), there is evidence that not all of the accommodative effort contributes to accommodative vergence. For example, barbiturates, amphetamines and ethanol influence the gain of accommodative vergence (AC/A ratio) but not the amplitude of the accommodative response (Westheimer, 1963; Hogan and Linfield, 1983). Latencies for accommodative vergence are 80–100 msec shorter than for accommodation (Wilson, 1973). The gain and phase of accommodative vergence differ from those of both accommodation and disparity vergence, regardless of whether the stimulus is a single sine wave or the sum of sine waves (Krishnan *et al.*, 1973). These differences could result from the innervation to accommodative vergence by a subset of sources for accommodation. Similarly, not all

sources of innervation for vergence may contribute to vergence accommodation.

In the current study, we have investigated possible innervational sources to crosslink interactions between accommodation and vergence by examining their dynamic interactions. Four topics have been addressed. The temporal frequency responses of accommodative vergence and vergence-accommodation expressed as gains (AC/A and CA/C ratios respectively) were measured in the first experiment in order to study possible phasic and tonic components of cross-link interactions between accommodation and vergence. The second experiment examined the contribution of feedback to the dynamic interactions between accommodation and vergence by measuring their temporal frequency responses while feedback to the stimulated system was cancelled. A third group of experiments examined the linearity of cross-link interactions as a function of stimulus amplitude. Responses to step stimuli, sinusoids and pseudo-random sinusoids were examined. The fourth group of experiments investigated whether accommodative vergence could cause adaptation of tonic vergence (change in phoria) and whether vergence accommodation could cause adaptation of tonic accommodation (changes in the resting focus). The results of these studies indicate that cross-link interactions depend upon both stimulus velocity and stimulus amplitude and that accommodation and vergence can be adapted directly to vergence accommodation and accommodative vergence respectively.

METHODS

Visual stimulus

The target was a one-dimensional difference of Gaussian (DOG), which resembled a bright, blurred, vertical bar with a dark, blurred, vertical bar on either side. The DOG was obtained by subtracting a broad Gaussian, luminance profile from a narrow one (Wilson, 1978), using the formula

$$\mathrm{DOG}(x) = 3\exp(-x^2/\sigma^2) - 2\exp(-x^2/2.25\sigma^2) + K$$

in which σ^2 is the space constant and K equals mean luminance. This spatially bandpass-filtered pattern was presented on an oscilloscope screen, which was photographed so that two identical, 35-mm color slides could be produced (one for each eye). Slides were needed because oscilloscope screens were too large to be compatible with the optometer that was used for stimulus presentation. Luminance profiles, obtained from the slides with a scanning photometer, were used to calculate contrast, using the formula

$$C = (L_{max} - L_{mean})/L_{mean}.$$

The contrast of the DOG, obtained in this manner, was 100%. Fourier transforms of the luminance profiles revealed a full bandwidth at half height of 1.75 octaves and a peak spatial frequency of 6.4 c/deg, which were the same as the mathematically-predicted bandwidth and spatial frequency (Schor *et al.*, 1984). The mean luminance was 10 cd/m^2.

Apparatus

Two serially-arranged, opto-electronic systems, one for stimulus presentation and the other for response measurement, were used (Fig. 1). The stimulus-presentation system was capable of independent control of the inputs to accommodation and vergence and of rendering those inputs either statically or dynamically, and under either closed-loop or open-loop conditions. The response-measurement system was capable of continuous and simultaneous monitoring of the outputs of accommodation and vergence and of producing high-resolution recordings.

The stimulus to accommodation was generated on a stepping-motor-operated, Badal optometer (Crane and Cornsweet, 1970) and the accommodative response was measured on a dynamic, infrared optometer (Knoll and Mohrman, 1972). The Badal optical system avoided proximal stimulation of accommodation by maintaining a constant retinal image size as the blur stimulus to accommodation was varied. The problem of eye-movement artifacts, which is common to dynamic infrared optometers in general (Krishnan *et al.*, 1977), was exacerbated during the measurement of dynamic vergence accommodation, during which lateral, eye movements were purposefully evoked in order to study their effects on the accommodative response. The subjects were trained to hold the right (measured) eye stationary to within 0.5 deg. To verify that the eye before the dynamic infrared optometer did indeed remain stationary, a 4-channel, strip-chart recorder was used, in which the movements of each eye were monitored separately, while on

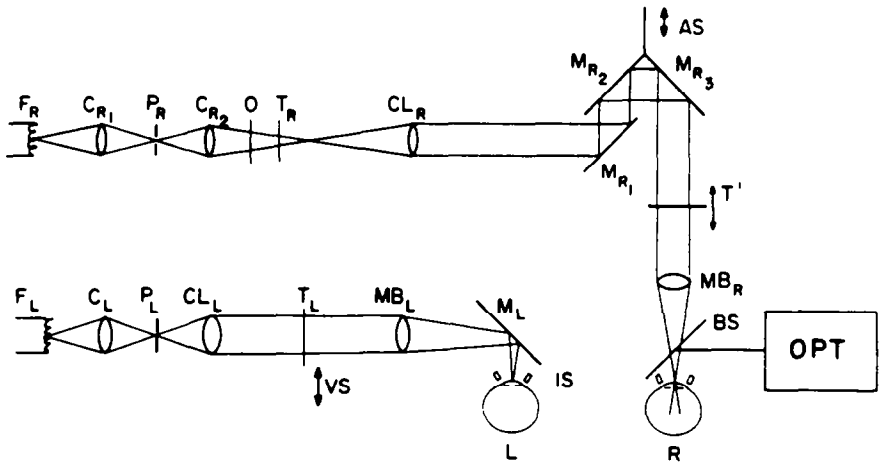

Fig. 1. A top view schematic illustrates the vergence-accommodation stimulator. Optical channels for the right and left eyes are both Badal stimulus optometers that can present stimuli in Maxwellian view (open-loop). Each channel has a filament light source that is imaged at a pinhole pupil and then eventually at the entrance pupil of each eye. The pinholes P_L and P_R serve as 0.75 mm aperture stops which, when imaged in the 6 mm eye pupil, increase the depth of focus to ± 4 D. Targets in the right channel (Tr) are imaged before the Maxwellian and Badal lens (MB_R) at T'. The image distance from MB_R determines the accommodative stimulus (AS) which is controlled by a stepping motor that moves mirrors M_{R_2} and M_{R_3}. The accommodative loop of the right eye can be closed by inserting a diffusing screen or opal glass (O) behind the target plane (T_R). Asymmetric vergence is stimulated (VS) by lateral motion of the left channel target (T_L) which is controlled by an X–Y plotter. Accommodative responses are measured objectively by the optometer (OPT) which receives infrared light reflected off the fundus of the right eye from a beam splitter (BS). Vergence responses are measured objectively by a pair of infrared sensors (IS) that sense light reflected from the front surface of each eye. F = filament; P = pinhole; C = condenser; CL = collimating lens; O = opal glass; T = target plane; MB = Maxwellian and Badal lens; M = mirror; AS = dynamic accommodative stimulator; VS = dynamic vergence stimulator; BS = beam splitter; OPT = optometer(Ophthalmatron); subscripts L and R = left and right eye respectively.

the other two channels accommodation and vergence were also recorded.

The vergence stimulus was provided by a function generator, which drove an X-Y plotter. The latter device was physically attached to the target slide before the left eye by means of a mechanical linkage, the movements of which adjusted the lateral position of the slide. Thus, an asymmetric vergence demand was generated without affecting the optical distance of the stimulus and without affecting the position of the right eye (which for the sake of optometer accuracy was required to remain stationary). Normally, asymmetric vergence involves both version and vergence components (Westheimer and Mitchell, 1956; Rashbass and Westheimer, 1961; Ono, 1983). Addition of these innervation sources is thought to occur at the muscle-globe plant (Kenyon *et al.*, 1980). This implies that combined version and vergence movements are processed independently. Consequently, the vergence innervation is predicted to have the same influence upon open-loop accommodation and disparity vergence dynamics during symmetric and asymmetric vergence stimulation.

The eye movements were measured with a Gulf-Western, model-200, infrared, eye-movement monitor. The sensor assembly was mounted on a pair of spectacle frames which were worn by the subjects. The infrared sources and the photodiodes were carefully adjusted to avoid crosstalk with the infrared optometer. The vergence signal was derived from the lateral movements of each eye by means of an independent differential amplifier, which had the effect of cancelling out conjugate eye movements.

Experimental procedures

The pupils were dilated to 6 mm with 2.5% phenylephrine, a weak sympathomimetic agent, in order to avoid optometer artifacts. The use of a much stronger dosage of phenylephrine (10%) has been shown to have no effect on the resting focus of accommodation (Garner *et al.*, 1983). A headrest and a mouthbite were also used for artifact abatement.

The accommodative loop was opened in two different ways: The first method used a Maxwellian view, which projected a pinhole

pupil (0.75 mm) into the plane of the subject's entrance pupil (Fig. 1). Although diffraction limits of the retinal image are greater with an artificial pinhole pupil placed close to the eye than with a projected "Maxwellian" pinhole pupil, for an eye with a large natural pupil (Westheimer, 1966), both pinhole pupils increase the eye's depth of focus. In this study the depth of focus for perception of blur increased from ± 0.35D in the normal condition to ± 4D in the Maxwellian view condition.

In the second open-loop method, the output of the dynamic, infrared optometer was fed back into the motorized, stimulus optometer (thus making the accommodative error independent of the amount of accommodation). With the accommodative loop opened, the sinusoidal variations of optical vergence were seen by the observer; however, accommodative responses had no influence upon the defocus of the retinal image. In the normal closed-loop state, the accommodative response reduces the error signal produced by a given change in target distance, so that changes in retinal image defocus are smaller than corresponding changes in target distance. In the second open-loop condition, retinal image blur is totally dependent upon target distance since the effects of accommodative responses have been cancelled. When the loop was opened, accommodation went to its resting focus. This resulted in a midpoint of sinusoidal oscillations at the subject's resting focus (1.37 D for J.K. and 1.50 D for M.N.). Subjects were instructed that although it might not be possible to always keep the target clear, they should attempt to focus it "naturally" without straining. Both subjects were able to perform this task easily on their first attempt, without any training or practice, as demonstrated by the small phase lags of accommodative vergence, compared to stimulus phase, plotted in Fig. 4.

During positive feedback, the accommodative system received the same odd error information (e.g. chromatic aberration) as it did under normal closed-loop conditions so that accommodation could respond in an appropriate direction that would normally reduce retinal image blur (Fincham, 1951). The availability of odd error information was shown in a similar experimental paradigm by Campbell and Westheimer (1959) who observed, with a hand driven stimulus optometer, that a cyclopleged observer could discriminate between myopia and hypermetropia.

The stimulus to accommodation used in the current study under the second open-loop condition was smaller than the stimulus used in the closed-loop condition in order to have similar amounts of blur at low temporal frequencies under the two conditions. It was also chosen to fall within the 1.5 D range in which accommodation can discriminate between positive and negative defocus (Fincham and Walton, 1957). The stimulus amplitudes chosen for the second open-loop condition resulted in similar response amplitudes at low temporal frequencies as found with the larger stimuli used in the normal closed-loop stimulus condition (Tables 1 and 2).

The vergence loop was also opened in two ways: the first was by extinguishing the light source before the subject's left eye and the second was by feeding the output of the eye-movement monitors back into the X–Y plotter, which supplied an asymmetric, vergence demand to the left eye (thus making the vergence error independent of the amount of vergence response). The small stimulus amplitudes used in the second open-loop condition resulted in similar retinal image disparities as presented by low temporal frequency stimuli under the normal closed-loop condition.

The CA/C frequency response was always measured with the accommodative loop opened, so that any change in accommodation could be attributed to a change in vergence. A binocular Maxwellian view was used to open the accommodative loops of both eyes. The vergence stimulus was either a temporally-modulated sine wave or a complex waveform obtained by summing three sine waves. The CA/C frequency response was obtained under two conditions: in the first the vergence loop was closed (referred to as the "closed-loop" condition) and in the second the vergence loop was opened by feedback of the output of the eye-movement monitors (referred to as the "open-loop" condition) (Rashbass and Westheimer, 1961). The CA/C ratio was defined as the change in accommodation in diopters divided by the change in vergence in meter angles (the response CA/C ratio).

The AC/A frequency response was always measured with the vergence loop opened, so that any change in vergence could be attributed to a change in accommodation. The vergence loop was opened by extinguishing the light source of the left eye. The accommodative stimulus was either a temporally-modulated sine wave or a complex waveform obtained by sum-

Table 1. Measures of accommodation and convergence during stimulation of accommodative vergence (top) and vergence accommodation (bottom) are shown for subject J.K.

AC/A measures (subject J.K.)								
Temporal	Closed loop		Open loop		Simple		Complex	
frequency (Hz)	A (2D)	V	A(1D)	V	A(1D)	V	A(D)	V
0.025	1.00	0.5	1.1	13.8	0.5	0.37	0.5	0.9
0.05	1.00	1.0	0.88	9	0.5	0.44	0.5	0.93
0.1	0.92	3.66	0.65	4.7	0.5	0.63	0.45	1.35
0.25	0.8	3.5	0.5	3.2	0.4	1.0	0.38	1.32
0.5	0.64	4.9	0.4	2	0.3	1.25	0.32	1.68
0.75	0.5	2.6	0.25	1.15	0.25	0.75	0.23	0.80
1.0	0.3	1.4	0.15	0.6	0.17	0.44	0.16	0.46
1.5	0.2	0.8	0.1	0.32	0.12	0.3	0.12	0.3
2.0	0.17	0.4	0.08	0.2	0.09	0.18	0.08	0.16
CA/C measures (subject J.K.)								
Temporal	Closed loop		Open loop		Simple		Complex	
frequency (Hz)	V(10Δ)	A	V(2Δ)	A	V(5Δ)	A	V(5Δ)	A
0.025	10	0.12	9.5	1.58	5	0.06	5	0.26
0.05	10	0.17	6	0.63	5	0.08	5	0.23
0.1	10	0.2	4	0.32	5	0.13	5	0.23
0.25	10	0.32	2.6	0.14	5	0.18	5	0.21
0.5	9.8	0.41	2	0.08	4.75	0.25	4.8	0.25
0.75	7.6	0.4	1.5	0.08	3.9	0.33	4	0.31
1.0	5.6	0.51	1.1	0.095	2.8	0.35	2.75	0.32
1.5	2.25	0.38	0.45	0.075	1.3	0.19	1.25	0.21

The stimulus amplitude is noted at the top of each data column and may be used to calculate stimulus AC/A and CA/A ratios. However, both motor responses have been plotted as ratios (response AC/A and response CA/C) in Figs 4–6. Note that the CA/C ratios plotted in Figs 5 and 6 are expressed in diopters/ meter angle (D/MA) whereas the data in the table is expressed in prism diopters (6 PD = 1 MA). Accordingly, CA/C ratios obtained from the bottom table must be multiplied by 6 before plotting them in Figs 5 and 6.

ming three sine waves. The AC/A frequency response was obtained under two conditions: In the first the accommodative loop was closed (referred to as the "closed-loop" condition) and in the second the accommodative loop was opened by feeding back the output of the dynamic, infrared optometer (referred to as the "open-loop" condition). The AC/A ratio was defined in vergence in prism diopters divided by the change in accommodation in diopters (the response AC/A ratio).

Table 2. Measures of accommodation and vergence during stimulation of accommodative vergence (top) and vergence accommodation (bottom) are shown for subject M.N. These data are plotted as described for Table 1 in Figs 4 and 5

AC/A measures (subject M.N.)				
Temporal	Closed loop		Open loop	
frequency (Hz)	A(2D)	V	A(1D)	V
0.025	1.5	6	2	19.4
0.05	1.4	7.13	1.5	13.27
0.1	1.5	10.5	1.0	10.63
0.25	1.2	8.3	0.75	7.27
0.5	0.9	4.72	0.55	3.5
0.75	0.75	3.17	0.4	2
1.0	0.45	1.68	0.22	0.9
1.5	0.38	0.95	0.15	0.3
CA/C measures (subject M.N.)				
Temporal	Closed loop		Open loop	
frequency (Hz)	V(10Δ)	A	V(2Δ)	A
0.05	10	0.4	9	0.9
0.1	10	0.57	5.75	0.54
0.25	10	0.66	3.2	0.34
0.5	9	0.86	1.85	0.23
0.75	7.5	0.89	1.5	0.21
1.0	5	0.67	1.0	0.14
1.5	2.5	0.60	0.5	0.10

Voltage analogs of the accommodative and vergence responses were run through a lowpass, analog filter, which had a −3 dB rolloff at 15 Hz. The filtered outputs were recorded on magnetic tape with an F.M. deck, using two high-resolution channels (with center carrier frequency of 4000 Hz) and simultaneously on a 4-channel, strip-chart recorder. The magnetic tape recordings were used to do FFTs so that the accommodative and vergence amplitudes could be precisely resolved at the stimulus frequency. The tape recordings were played back into an oscilloscope to inspect for artifacts (e.g. blinks or eye movements), which were recognizable as a sudden saturation of the signal. Artifact-free segments were digitized by an A-D converter at a rate of 10 samples/sec. Typically, 20 sec of data were analyzed, except at low temporal frequencies where 40 sec were used.

Subjects

Two subjects (J.K. and M.N.) with normal AC/A and CA/C ratios were studied in all four groups of experiments. Two additional subjects, one with a low CA/A ratio (J.T.) and one with a low AC/A ratio (M.W.) were studied in the last two groups of experiments in which linearity and adaptation were investigated. All four subjects had normal eye alignment and 20 sec arc stereoacuity.

RESULTS

Closed-loop frequency response

Figures 2 and 3 show typical responses of "closed-loop" accommodative vergence and vergence accommodation respectively to low frequency (0.06 and 0.05 Hz) and moderately high frequency (0.25 Hz) single sinusoidal stimuli. It is clear that both the AC/A and CA/C are lower with low than high frequency stimuli. The change in gain with temporal frequency is

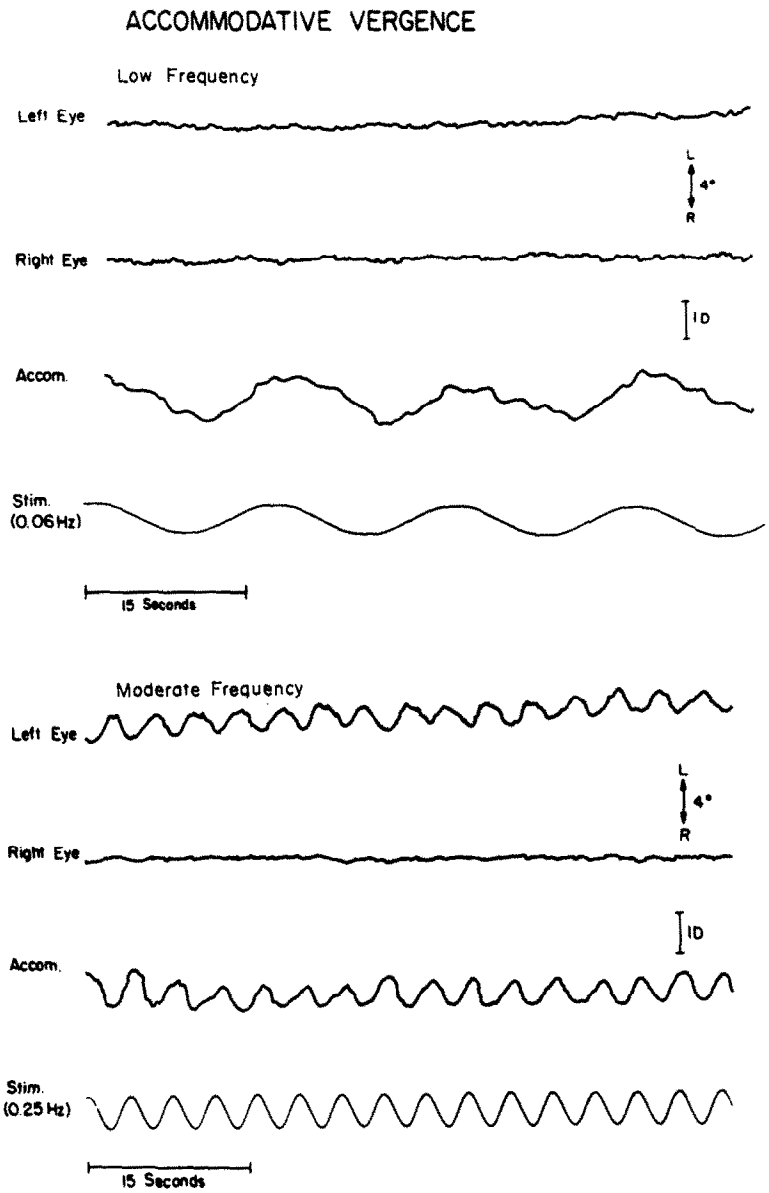

Fig. 2. Accommodation was stimulated in the right eye while the left eye was occluded. Movements of the left eye were stimulated by accommodative vergence. Shown are the accommodative vergence responses to low (0.06 Hz) and moderately high (0.25 Hz) frequency sinusoidal blur stimuli presented under "closed-loop" conditions. Accommodative vergence was reduced with the low temporal frequency stimulus.

not due to differences in accommodative gain for the AC/A or vergence gain for the CA/C, since amplitudes of accommodation and vergence responses to the two frequencies are unchanged (see Tables 1 and 2).

The solid curves in Figs 4 and 5 plot the "closed-loop" gain and phase of accommodative vergence and vergence accommodation responses for both subjects to a range of temporal frequencies (0.025–1.5 Hz). Stimulus amplitudes for accommodation and vergence were 2 D (peak to peak) and 10 PD (peak to peak) respectively. There is a low-frequency rolloff of both the response AC/A and CA/C ratios. The low frequency rolloff is similar on the log–log coordinates for the two ratios; however, the CA/C continues to rise as temporal frequency increases. A comparison was made between these frequency response data and measures of the step response of accommodative vergence and vergence accommodation. The AC/A ratios in response to a 2 D step were

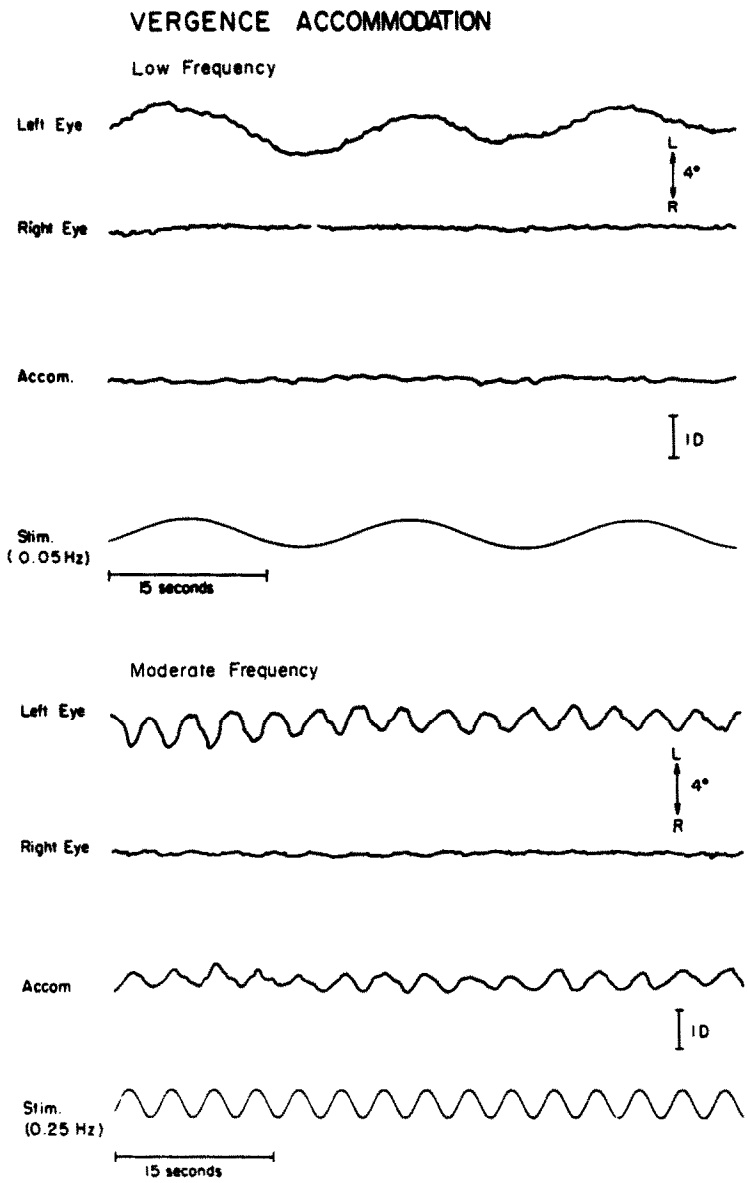

Fig. 3. Disparity vergence was stimulated with an asymmetric disparity presented to the left eye in order to allow accommodation to be monitored in the right eye. The accommodative loop was opened by a 0.75 mm pinhole aperture stop imaged in the eye pupil plane in Maxwellian view. Shown are the vergence accommodation responses to low (0.05 Hz) and moderately high (0.25 Hz) frequency of sinusoidal disparity oscillations presented under "closed-loop" conditions. Vergence accommodation was reduced with the low temporal frequency stimulus.

5 PD/D for subject J.K. and 5.7 PD/D for subject M.N. The CA/C ratios in response to a 10 PD step were 0.55 D/MA for subject J.K. and 0.45 D/MA for subject M.N. These step response AC/A and CA/C ratios equaled amplitudes of AC/A and CA/C ratios shown in Figs 4 and 5 stimulated at temporal frequencies ranging from 0.3 to 0.6 Hz where the gains of the closed-loop responses were highest (see Tables 1 and 2).

At high temporal frequencies both accommodation and resulting accommodative vergence were reduced in amplitude (see Tables 1 and 2). Accommodative vergence was reduced slightly more than accommodation such that the ratio of these two responses (AC/A) was attenuated slightly at high temporal frequencies. In contrast, the closed-loop response CA/C ratio of both subjects continued to increase as temporal frequency increased (see Tables 1 and 2). The response ratio increased because even though the responses of both motor systems were either reduced or remained constant at high temporal frequencies, disparity vergence was attenuated more than vergence accommodation. It is possible that under the difficult disparity vergence tracking conditions, experienced at high temporal frequencies, there was a recruitment of accommodative vergence similar to previous reports of an extension of disparity vergence amplitude by accommodative vergence (Westheimer and Mitchell, 1956; Morgan, 1968). It is also possible that there was greater fatigue or damping of the vergence mechanism than for accommodation at high temporal frequencies (Wilson, 1973).

Open-loop frequency response

The focus error was produced by stimulating accommodation with a 1 D peak to peak sinusoidal error summed with the accommodative responses monitored by a dynamic infrared optometer. This procedure cancelled the physiological negative feedback to accommodation with an equal but opposite positive feedback. Under these conditions the subject observed a sinusoidal variation in defocus ±0.5 D that was unaffected by attempts to clear

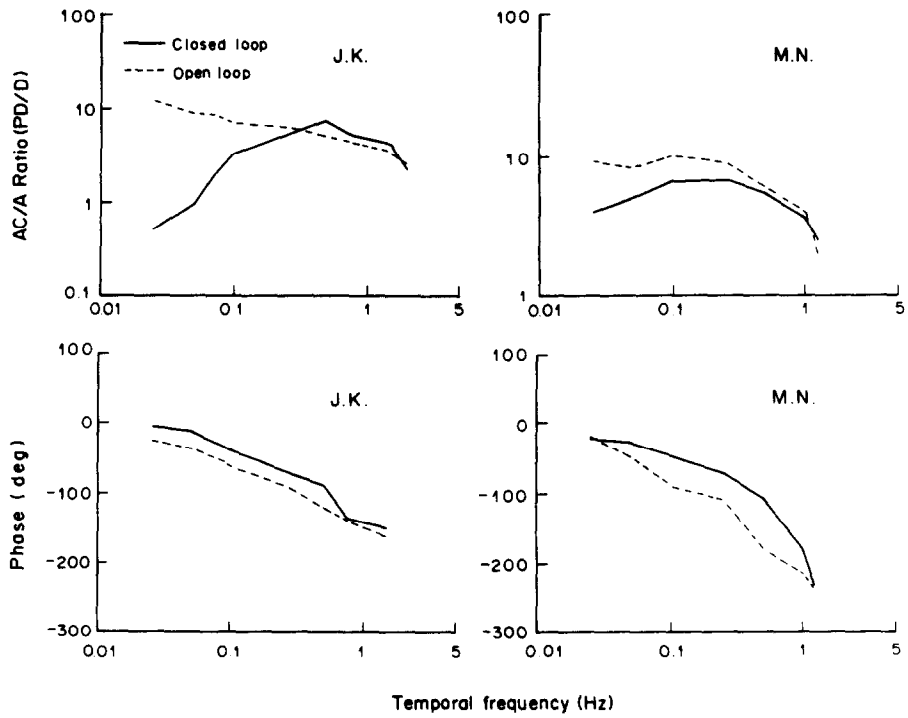

Fig. 4. Gain and phase plots, for two subjects, of accommodative vergence responses to sinusoidal blur stimuli presented under open-loop (dashed line) and closed-loop (solid line) conditions. The "closed-loop" condition is described in Fig. 2. The "open-loop" condition included feeding the output of the dynamic infrared optometer back to the motorized stimulus optometer, thus making the accommodative error independent of accommodative responses. Gain is the response AC/A ratio. Phase lag equaled the difference between accommodative vergence phase and stimulus phase. Stimulus amplitudes for open and closed-loop conditions were ±0.5 and ±1.0 D respectively. The low frequency rolloff in gain was only present under "closed-loop" conditions.

the image. This small defocus was similar to the amount of blur during 2 D low frequency closed-loop stimulation of accommodative vergence.

While measuring vergence accommodation, uncorrectable open-loop vergence error (retinal image disparity) was produced by stimulating disparity vergence with a 2 PD sinusoidal disparity variation and the vergence responses were monitored by an infrared eye-movement monitor. In this condition the subject observed a disparity that varied sinusoidally (± 1 prism diopter) from crossed to uncrossed, whose amplitude was unaffected by attempts to fuse the image. This small stimulus was similar to retinal disparities present during low frequency closed-loop stimulation of vergence accommodation.

The "open-loop" AC/A frequency-response (vergence response/accommodative response) and the "open-loop" CA/C frequency-response (accommodative response/vergence response) are plotted as dashed curves in Figs 4 and 5. There was no low-frequency rolloff of either accommodative vergence or vergence accommodation under "open-loop" conditions. At high temporal frequencies there was little difference between the open- and closed-loop responses. The AC/A ratio was highest during "open-loop" conditions at low temporal frequencies where it had been lowest during "closed-loop" conditions. Interestingly, neither accommodative responses to defocus nor vergence responses to disparity had low frequency rolloffs under either the open-loop or closed-loop stimulus conditions (see Tables 1 and 2).

Phase lags of vergence accommodation and accommodative vergence also varied under open-loop and closed-loop conditions (Figs 4 and 5). Phase lags were computed by comparing the phase of the response to phase of the stimulus. Both accommodative vergence and vergence accommodation show an increased phase lag at low temporal frequencies under the open-loop compared to closed-loop conditions. Under open-loop conditions, the phase lag increased 30–50 deg at low temporal frequencies (<0.1 Hz) for accommodative vergence and vergence accommodation, but there was no

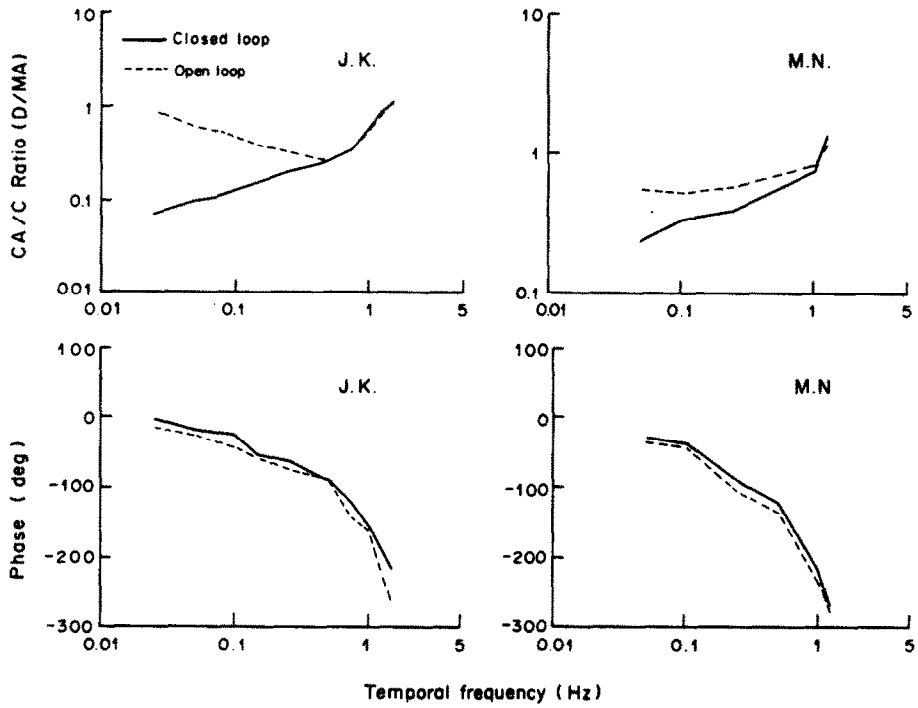

Fig. 5. Gain and phase plots for two subjects of vergence accommodation responses to sinusoidal disparity stimuli presented under open-loop (dashed line) and closed-loop (solid line) conditions. The "closed-loop"condition is described in Fig. 3. The "open-loop" condition included feeding the vergence output of the eye movement monitor back into the X–Y plotter which supplied an asymmetric vergence demand to the left eye, thus making the vergence error independent of the vergence response. Gain is the response CA/C ratio. Phase lag equaled the difference between vergence accommodation phase and stimulus phase. Stimulus amplitudes for open and closed-loop conditions were ± 1 prism diopters and ± 5 prism diopters respectively. The low frequency rolloff in gain is only present under "closed-loop" conditions.

phase difference between open-loop and closed-loop conditions at higher temporal frequencies. Similar differences in phase of open-loop and closed-loop accommodative vergence were observed by Brodkey and Stark (1967).

Linearity

Stimulus amplitude dependence of the AC/A and CA/C ratios was examined in subject J.K. by comparing responses to small and large stimuli at low (0.06 Hz) and high (0.25 Hz) frequencies. At the low frequencies, the response AC/A ratio increased from 2 to 5.6 PD/D when accommodation increased in response to stimulus amplitude from 0.87 to 2.25 D. At the higher stimulus frequency, the response AC/A ratio remained the same (5.4 PD/D) when the accommodative response increased from 0.75 to 2.00 D. Similarly, at a low temporal frequency (0.06 Hz) the CA/C ratio increased from 0.1 to 0.6 D/MA when the amplitude of the disparity vergence response increased from 5 to 10 PD. At higher temporal frequencies the CA/C was independent of the response amplitude of disparity vergence.

The amplitude dependent nonlinearity was also observed with complex temporal wave forms composed of the sum of 3 nonharmonic sinusoids. Three complex wave forms used were composed of 0.025, 0.06 and 0.1 Hz; 0.25, 0.45 and 0.8 Hz; and 0.06, 1 and 1.5 Hz. The stimulus amplitude of individual components was 1 D for accommodation and 5 PD for vergence. The amplitude of the complex waveform ranged from 2.5 to 3 times the amplitude of individual sub-components. The amplitude of motor responses to simple (single sinusoidal) and complex waveforms was calculated from the Fast Fourier Transform (FFT) that was performed with a micro Nova computer.

Frequency responses of the AC/A and CA/C obtained with simple and complex waveforms are illustrated in Fig. 6. The low frequency rolloff obtained with simple sinusoids is not present in the responses to the complex waveforms for either accommodative vergence or vergence accommodation. Responses at high frequencies to simple and complex waveforms are indistinguishable. Similar results were obtained for accommodative vergence by Krishnan *et al.* (1973) and for vergence accommodation by Yoshida and Watanabe (1969).

The amplitude dependent nonlinearity also expressed itself in the step response of crosslink

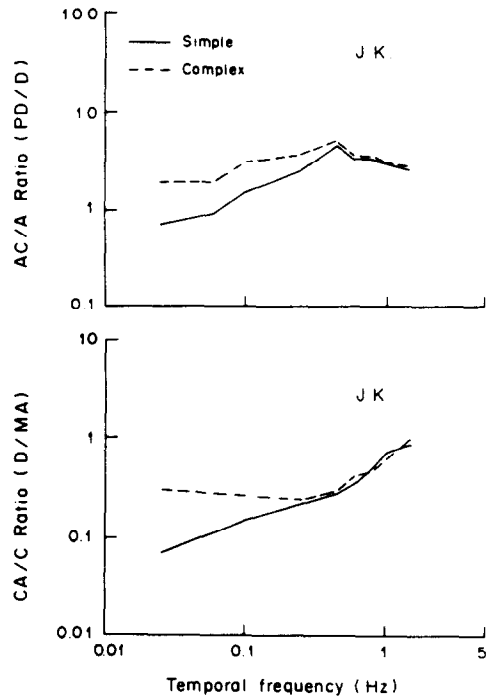

Fig. 6. Frequency response of the AC/A and CA/C ratios to single sinusoids (solid line) (1 D and 5 PD stim) and the sum of three nonharmonic sinusoids (dashed line) (1 D and 5 PD for each component). Both AC/A and CA/C ratios were higher at low frequencies in response to the complex than simple sinusoid.

interactions of two subjects who had either a low AC/A ratio (M.W.) or a low CA/C ratio (J.T.). Figure 7 illustrates for subject M.W., the step response of accommodative vergence to a small (1 D) (top) and large (2 D) (bottom) monocular stimulus to accommodation. During stimulation, the accommodative vergence response to the small step decayed rapidly (within 5 sec) while the large step response of accommodative vergence contained both a transient and sustained component. Figure 8 illustrates for subject J.T., the step response of vergence accommodation to a small (4 PD) (top) and a large (8 PD) (bottom) disparity stimulus while the accommodation loops were opened by pinhole pupils. The vergence accommodation response to the small step decayed rapidly whereas the larger step response had both a sustained and transient component. These transients of accommodative vergence and vergence accommodation were not observed in the other two subjects who had more normal (higher) AC/A (5 PD/D) and CA/C (0.55 D/PD) ratios.

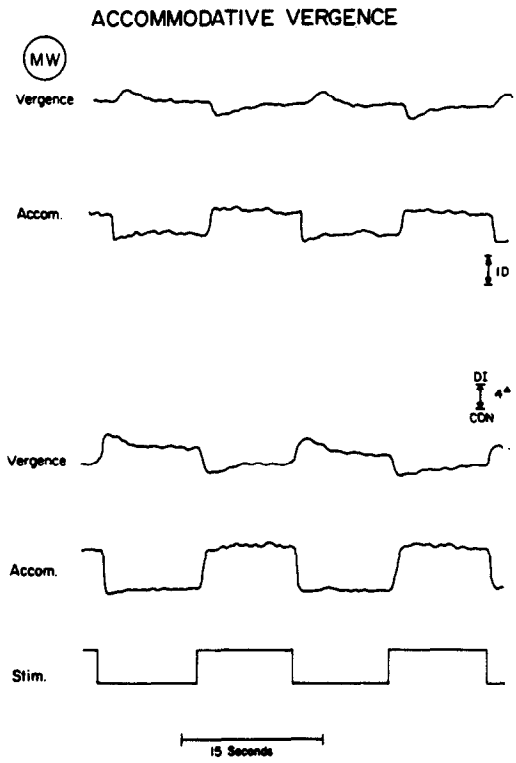

Fig. 7. Step responses of accommodative vergence to small (1 D) (top) and large (2 D) (bottom) monocular stimuli are shown for subject M.W. who had a low AC/A ratio (2.5 PD/D) and a high CA/C ratio (0.9 D/MA). During stimulation, the accommodative vergence response to the small step decayed rapidly (within 5 sec) while the large step response was composed of both a transient and sustained component.

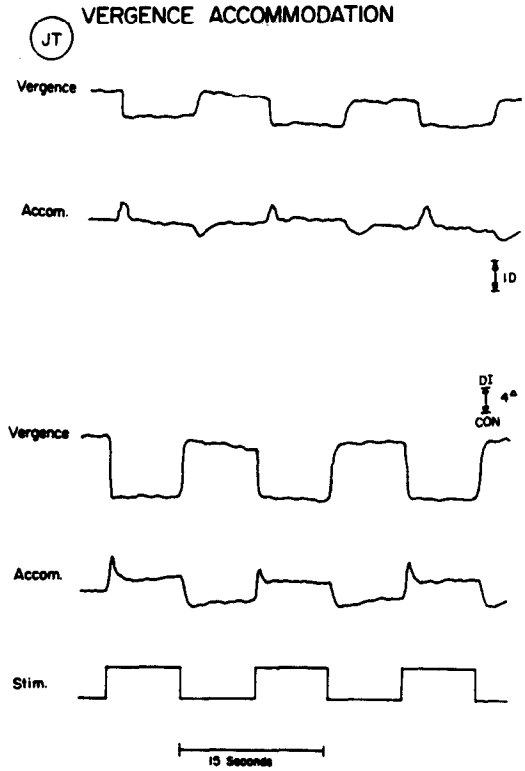

Fig. 8. Step responses of vergence accommodation to small (4 PD) (top) and large (8 PD) (bottom) disparities, while the accommodative loops were opened with pinholes, are shown for subject J.T. who had a low CA/C ratio (0.3 D/MA) and a high AC/A ratio (8 PD/D). During stimulation, vergence accommodation responses to small steps decayed rapidly (within 5 sec) while the larger step response was composed of a transient and sustained component.

Aftereffects of vergence accommodation and accommodative vergence

Adaptation of tonic vergence and tonic accommodation to cross link interactions was investigated by adapting one system (accommodation or vergence) to a step stimulus while the other system (vergence or accommodation respectively) was open-loop. Aftereffects of the adaptation stimulus were then measured simultaneously for both vergence and accommodation while they were both in open-loop states.

Aftereffects of vergence accommodation. Subjects binocularly fused a DOG pattern viewed through a binocular Maxwellian view (so that the accommodative loops of both eyes were opened). Vergence responses to 6–8 PD BO were stimulated for either 5 sec or 2 min and then the left eye was occluded by turning off its target. Both accommodation and vergence were monitored objectively during and after the adaptation period.

During adaptation to the BO prism, subject J.K.'s convergence response increased 8 PD and his accommodative response increased by 0.75 D. As expected, when the left eye was occluded after wearing the prism for a short period (5 sec), there was a rapid (5 sec) decay of both vergence and vergence accommodation. However, as illustrated in Fig. 9 (top), when the left eye was occluded after wearing the prism for 2 minutes, there was a residual esophoria (an increase of tonic vergence) and also a prolonged increase in resting focus (an increase of tonic accommodation). Both aftereffects usually decayed slowly over a 1–1.5 min period; however, in some subjects the esophoria decayed rapidly (within 5 sec) while the accommodative aftereffect persisted for over 1 min. This dissociation has been observed in subject M.W. who had a low AC/A ratio (2.5 PD/D) and who also adapted rapidly to accommodation (5 sec) but slowly to disparity vergence (in more than 1 min). This subject demonstrated a larger

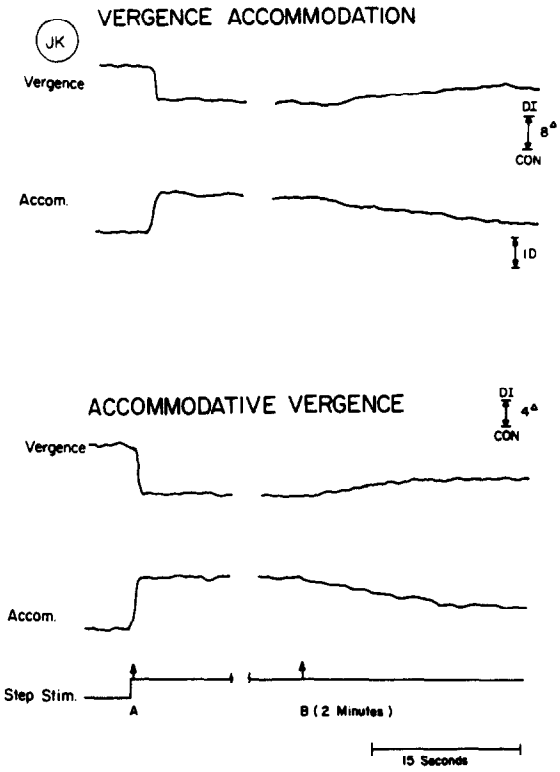

Fig. 9. Aftereffects of accommodation and vergence are shown for subject J.K. who had normal AC/A (5 PD/D) and CA/C (0.55 D/MA) ratios. Aftereffects in the upper half of the figure followed 2 min of adaptation to 8 base out prism while the accommodative loop was opened by pinhole pupils. After 2 min of adaptation, (B), the vergence loop was opened by occluding the left eye. Both vergence and accommodation decayed slowly with persistent aftereffects for both motor systems. Aftereffects, in the lower half of the figure, followed 2 min of monocular adaptation to a 2 D accommodative stimulus. The accommodative loop was opened at (B) with a pinhole pupil. Both vergence and accommodation decayed slowly with persistent aftereffects for both systems.

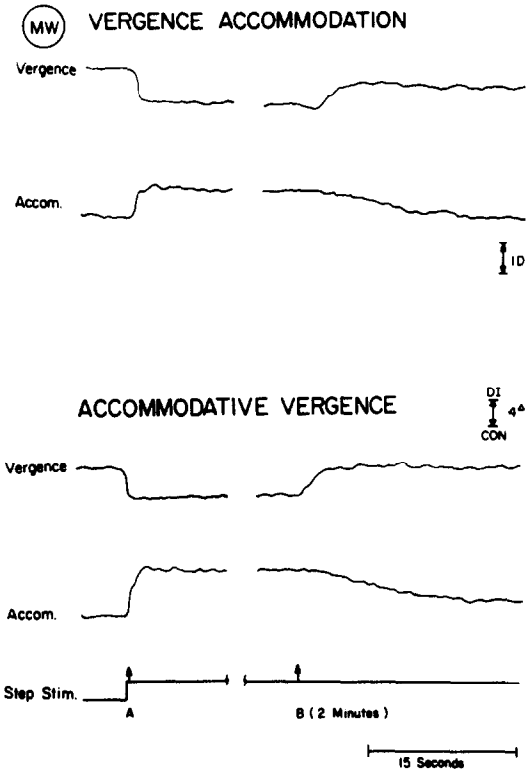

Fig. 10. Aftereffects of accommodation and vergence are shown for subject M.W. who had a low AC/A ratio (2.5 PD/D) and a high CA/C ratio (0.9 D/MA). Aftereffects shown in the upper half of the figure followed 2 min of adaptation to 6 base out prism while the accommodative loop was opened by pinhole pupils. After 2 min of adaptation, (B), the vergence loop was opened by occluding the left eye. The accommodative response persisted while vergence decayed rapidly. Aftereffects, in the lower half of the figure, followed 2 min of monocular adaptation to a 2 D accommodative stimulus. The accommodative loop was opened at (B) with a pinhole pupil. As it did following vergence adaptation, the accommodative response persisted while the vergence response decayed rapidly.

aftereffect of accommodation following adaptation to either a prism or lens (Fig. 10, top and bottom respectively). This dissociation of aftereffects indicates that both accommodation and vergence are adapting separately to the disparity vergence response and that the aftereffect of vergence accommodation is not the result of tonic vergence stimulating accommodation.

Aftereffects of accommodative vergence. Subjects accommodated to a 2 D stimulus presented only to the right eye for either 5 sec or 2 min and then the accommodative loop was opened using a Maxwellian view. Both accommodation and vergence were monitored during and after the adaptation period. During adaptation J.K. accommodated 1.5 D and had an accommodative vergence response of 7.5 PD. When the accommodative loop was opened after 5 sec of adaptation, there was a rapid (5 sec) decay of both accommodation and accommodative vergence. When the accommodative loop was opened after 2 min of adaptation, there was a residual increase of the resting focus (increase of tonic accommodation) as well as an esophoria (increase of tonic vergence) (Fig. 9, bottom). Both aftereffects usually decayed slowly over a 1–1.5 min period to their resting levels; however, as in the previous experiment, the two responses had markedly different time courses for subject J.T. who had a low CA/C ratio (0.3 D/MA) and who also adapted vergence rapidly to prism

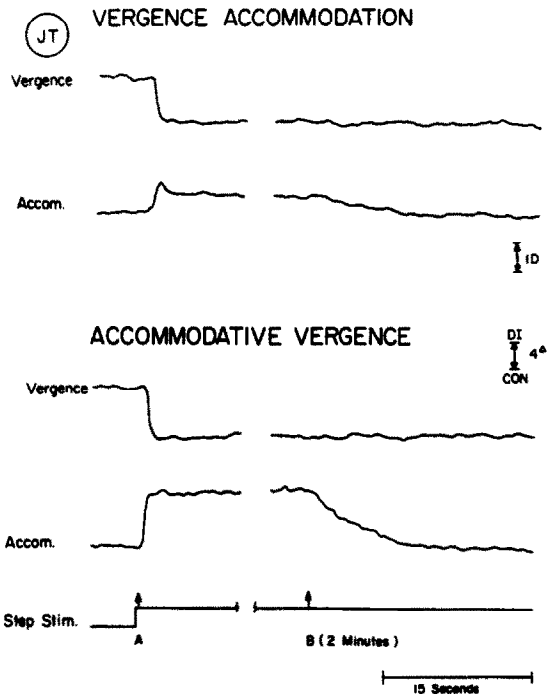

Fig. 11. Aftereffects of accommodation and vergence are shown for subject J.T. who had a low CA/C ratio (0.3 D/MA) and a high AC/A ratio (8 PD/D). Aftereffects shown in the upper half of the figure followed 2 min of adaptation to 8 base out prism while the accommodative loop was opened by pinhole pupils. After 2 min of adaptation, (B), the vergence loop was opened by occluding the left eye. The vergence response persisted while accommodation decayed rapidly. Aftereffects, in the lower half of the figure, followed 2 min of monocular adaptation to a 2 D accommodative stimulus. The accommodative loop was opened at (B) with a pinhole pupil. As it did following vergence adaptation, the vergence response persisted while the accommodative response decayed rapidly.

(5 sec) but adpated accommodation slowly to lenses (40–80 sec). This subject had greater aftereffects of vergence than of accommodation whether he was adapting to a prism or lens (Fig. 11, top and bottom respectively). This dissociation in the time-courses indicates that vergence and accommodation are adapting separately to reflex accommodation and that the aftereffect of accommodative vergence is not the result of tonic accommodation stimulating vergence.

DISCUSSION

Phasic control of crosslink interactions between accommodation and vergence

The low frequency rolloffs of the frequency responses of accommodative vergence and vergence accommodation measured under closed-loop conditions indicate there may be a velocity sensitivity for cross link interactions. Indeed, responses to small step stimuli of accommodative vergence and vergence accommodation decayed rapidly after an initial phasic response for subjects with low AC/A and low CA/C ratios respectively. The low AC/A and CA/C ratios were associated with robust adaptation of accommodation and vergence respectively. These results can be interpreted to mean that the postulated phasic response of accommodation, but not its tonic response, stimulates accommodative vergence, and that the postulated phasic component of disparity vergence, but not its tonic counterpart, stimulates vergence accommodation. The absence of the low frequency rolloff of accommodative vergence and vergence accommodation during open-loop sinusoidal stimulation can be interpreted to show a reciprocal relationship between tonic adaptation and the cross link interactions. Cancellation of feedback could have prevented tonic adaptation from reducing the stimulus to the postulated phasic element and its cross link innervation. For example, during the open-loop condition, tonic changes in the vergence phoria would not reduce the stimulus disparity presented to the proposed phasic disparity vergence mechanism. Consequently, the proposed phasic response to low temporal frequencies could continue to stimulate vergence accommodation. When feedback was not cancelled, low frequency variations in disparity could have been negated by vergence adaptation. This adaptation could allow both phasic disparity vergence and its stimulus to vergence accommodation to become reduced. A similar reciprocal relationship between adaptable tonic accommodation and accommodative vergence would account for the stimulus velocity dependence of the AC/A ratio.

Linearity

The velocity sensitivity of cross links between accommodation and vergence had a nonlinear amplitude dependence. Both the AC/A and CA/C ratios increased with stimulus amplitude. This nonlinearity can be modeled as a low saturation limit or amplitude window for stimuli to which vergence and accommodation adapt. Small stimuli within the window could be adapted to completely and, following the discussion above, there would be little or no cross link interaction. Stimuli larger than the window

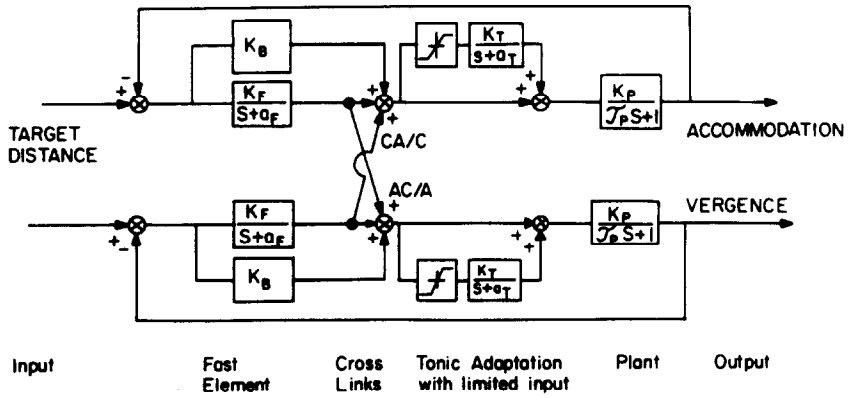

Fig. 12. System model representation of synchenetic interactions between accomodation and vergence. It is proposed that two motor systems are interconnected at points in their feed forward paths located between proposed phasic and tonic neural integrators. The input to tonic integrators has a proposed saturation limit or stimulus window which could produce amplitude dependent nonlinearities of the AC/A and CA/C ratios.

would be adapted to partially. The proposed phasic mechanism and its cross link could respond to the remaining unadapted portion of the stimulus. The hypothesized stimulus window for adaptation may account for previously reported amplitude dependent nonlinearities for the AC/A ratio (Flom, 1960; Alpern *et al.*, 1959; Ogle and Martens, 1957; Brodkey and Stark 1967).

Aftereffects of accommodative vergence and vergence accommodation

The results of the current study illustrate that not only does tonic vergence adapt to disparity vergence but it also adapts to accommodative vergence. Similarly, tonic aftereffects of accommodation result from both accommodative stimulation (Schor *et al.*, 1984) and from vergence accommodation. Prior studies have not revealed aftereffects of accommodative vergence on the phoria or aftereffects of vergence accommodation on the resting focus of accommodation. Previous measures of accommodative-vergence aftereffects used shorter adaptation periods and objectively monitored accommodative vergence but not accommodation (Schor, 1979). Prior measured of vergence-accommodation aftereffects were made in darkness after adapting to combined prism and lens stimuli (Owens and Leibowitz, 1980). No aftereffect would be expected for accommodation under these conditions because tonic aftereffects of accommodation are masked in darkness (Schor *et al.*, 1986). In addition, it would be difficult to separate the combined effects of the lens from those of the prism.

A heuristic model of dynamic interactions between accommodation and vergence

The results of this study are summarized by a model developed previously to illustrate adaptation of vergence (Schor, 1979) and accommodation (Schor *et al.*, 1984). The model is shown in Fig. 12. and it is meant to provide a parsimonious description of dynamic interactions between accommodation and vergence. Physiological correlates have been observed in the monkey midbrain for phasic (burst neurones) and tonic pre-motoneurones that control vergence eye movements (Mays, 1984; Tello and Mays, 1984). However the physiological organization for the vergence control and its interaction with accommodation are still under investigation. The salient features of the model include a phasic-tonic control mechanism modeled here for both accommodation and vergence as leaky neural integrators with short and long time constants. Cross links between the motor systems are shown to receive their input from the postulated phasic controller of one system, and to provide input to the tonic controller of the other system. The proposed tonic elements are shown to have limited amplitudes or windows for their stimuli. There is also a parallel phasic controller (Kb) that is similar to one proposed by Krishnan and Stark (1977) to account for the rapid phasic response of accommodation and vergence to abrupt stimuli.

The model provides a graphic summary of our observations of dynamic interactions between accommodation and vergence. The model describes the actions of accommodative vergence and vergence accommodation as comp-

lementary to the actions of tonic vergence and postulated tonic accommodation. The tonic elements would be sufficient to sustain the motor responses which were initiated by the fast but leaky neural integrators. During sustained responses, the contribution of the fast integrators would gradually be reduced as the contribution of the tonic integrators increased. Therefore, accommodative vergence and vergence accommodation, which are presumed to be controlled by the fast integrators, would become dormant during the sustained responses. In contrast, during transient responses the contribution of the proposed fast integrators would be substantial, which would allow accommodative vergence and vergence accommodation to play a more dominant role. This organization could serve to reduce individual response efforts of accommodation and vergence to abrupt stimuli because control would be efficiently shared by the two motor systems.

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